

Week 6 Baseline Triage Model Report

To: Dr. L. De Freitas, Dr. Marcus Reyes & Mercer General Hospital ED Board
From: Mithra Ramoutar, Clinical AI Research Trainee
Re: Baseline logistic regression and decision tree for ESI prediction , Yale EMLC dataset

1. Recap: what we're predicting and on what data

We are predicting the Emergency Severity Index (ESI, levels 1–5) from information available at triage , vitals and chief-complaint flags only. As documented in the Week 5 feasibility memo, two columns (disposition, previousdispo) are excluded from every model because they are only known *after* triage; including them would leak the answer into the input. The row index column (Unnamed: 0) was also excluded , it carries no clinical information and was flagged for removal in the Week 5 cleaning log.

The dataset remains severely imbalanced: ESI 1 makes up 0.1% of encounters (77 of 55,121; only 16 land in the 20% test split). That imbalance drives every result below , a model can look "accurate" while missing almost all of the sickest patients.

2. Models

Two baselines were trained on an 80/20 stratified split (random_state=42, same seed used in both notebooks so results are reproducible):

- **Logistic regression** , features scaled with StandardScaler, max_iter=1000.
- **Decision tree** , max_depth=5, unscaled features (trees split on raw thresholds, so scaling doesn't change the result).

Both were benchmarked against a DummyClassifier(strategy="stratified"), which guesses ESI levels in the same proportion they appear in the data , the floor any real model must clear.

3. Benchmark table

		Weighted F1	
Model Accuracy Macro	F1	Recall (ESI 1)	
		Recall (ESI 1)	
		Recall (ESI 1)	

Dummy (baseline)

Logistic Regression

Decision Tree 0.556 0.216 0.463 **0.000**

Both real models beat the dummy on accuracy and macro F1. But the column that matters , recall on ESI 1 , tells a much less comforting story: logistic regression catches 1 in 4 of the sickest patients; the decision tree at this depth catches **none of them**.

4. Primary metric: recall for ESI 1

We are choosing recall on ESI 1 as our primary metric, not accuracy and not weighted F1.

Accuracy is dominated by ESI 3, which is 49% of the dataset. The decision tree makes this concrete: it reaches 55.6% accuracy , beating the dummy by 18 points , while never once predicting ESI 1, ESI 4, or ESI 5 on the test set (see Section 5). Weighted F1 (0.463 for the tree) has the same blind spot, since it's dominated by the same large classes. Macro F1 is more honest , it drops the tree to 0.216, which is *below* the dummy's 0.204 in the metric that actually treats the classes fairly , but it still blends four other classes' scores into ESI 1's, so it can't answer the one question we need answered.

Recall for ESI 1 answers that question directly: **of the patients who were truly critical, what share did the model catch?** A false negative here is the costliest mistake in the whole project , a missed ESI 1 means a resuscitation-level patient was told to wait.

On macro vs. weighted F1, concretely: weighted F1 averages each class's F1 score weighted by how many patients are in it, so ESI 3 (49% of patients) dominates the number and a model can score well by only getting the common classes right. Macro F1 gives every class , including the 16 ESI 1 patients in the test set , equal weight in the average, so failing on the rare, critical class pulls the score down sharply. That's why the decision tree's weighted F1 (0.463) looks so much healthier than its macro F1 (0.216): the weighted number is hiding the exact failure we care about.

5. Failure mode

Which group does the model miss most? ESI 1 patients, for both models , but the tree fails completely (0/16 caught) while logistic regression catches 4/16.

Looking at where the true ESI 1 test patients actually landed:

- **Logistic regression:** 4 correctly called ESI 1; 10 called ESI 2; 1 called ESI 3; 1 called ESI 4. Most of its misses are "near misses" (adjacent classes), not wild ones.
- **Decision tree:** 8 called ESI 2, 8 called ESI 3 , zero caught, and it never predicts ESI 1 anywhere in the test set at all.

Why? With only 77 ESI 1 encounters in the full dataset (about 61 in the 80% training split), both models see far too few examples to learn what separates a resuscitate-now patient from an emergent one. At `max_depth=5`, the tree's splits are governed by the classes with the most data (ESI 2 and 3), so it never carves out a branch fine enough to isolate ESI 1 at all , it effectively treats the rare class as noise.

So what? A critically ill patient triaged as ESI 2 or 3 would wait behind less urgent cases. For the tree specifically, that isn't a rare failure , it is the guaranteed outcome for every ESI 1 patient at this depth.

Next step: class-weighting or resampling so the rare, critical class gets real weight during training (see below), and comparing recall for ESI 1 before and after , not just accuracy.

6. Next steps (Week 7)

- Class-weighting (`class_weight="balanced"`) or resampling (SMOTE) to give ESI 1 more signal during training, as flagged in the Week 5 memo's Concern 1 mitigation. Given the tree's 0% recall here, this is the highest-priority fix.
- Sweep the tree's `max_depth` to check whether more depth recovers ESI 1 recall without overfitting the test set , 0/16 suggests depth 5 is too shallow to isolate the rare class at all, independent of overfitting risk.
- Local validation against Mercer General data before any of these figures are treated as more than a US academic-hospital benchmark (Week 5 memo, Reason 3 / Caveat).

Reproducibility: random_state=42 throughout; dataset is yaleemmlc_admissionprediction_triage.csv (55,121 encounters, 226 columns), with disposition, previousdispo, and the row-index column excluded from all model inputs.